

# Project Design Phase

## Proposed Solution Template

|               |                                                  |
|---------------|--------------------------------------------------|
| Date          | 03 October 2026                                  |
| Team ID       | 6ab7f10630f9767f87728ca3                         |
| Project Name  | Import Data Using Transform Maps (Spreadsheet)   |
| Team Leader   | E Dilli Kumar                                    |
| Team Members  | R Vinish, K Jerina, B Mohanavalli, P Sri Ragavan |
| Maximum Marks | 2 Marks                                          |

### Proposed Solution Template:

The project team has filled in the following information in the proposed solution template.

| S.No. | Parameter                                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.    | Problem Statement<br>(Problem to be solved) | Organizations keep large amounts of data in spreadsheets (Excel / CSV), and entering it into the system by hand is slow, repetitive and error-prone. Manual entry causes duplicate records, wrong field values and missing data, and users cannot easily tell which rows failed. Users need a quick and reliable way to load spreadsheet data into the correct tables without coding.                                                                                                                                                                                                                                |
| 2.    | Idea / Solution<br>description              | <ul style="list-style-type: none"><li>• The user uploads a spreadsheet (.xlsx / .csv) through a Data Source, which loads the rows into an Import Set (staging) table.</li><li>• A <b>Transform Map</b> maps each source column to a field of the target table, and <b>coalesce</b> on key fields updates existing records instead of creating duplicates.</li><li>• Transform scripts convert data types, clean values and resolve reference fields.</li><li>• The transform inserts or updates the records in the target table, and an import log reports the inserted, updated, skipped and failed rows.</li></ul> |
| 3.    | Novelty / Uniqueness                        | <ul style="list-style-type: none"><li>• Codeless, mapping-based import: users configure field maps instead of writing scripts.</li><li>• Reusable Transform Maps that can be run again for repeated or scheduled imports.</li><li>• Built-in duplicate prevention through coalesce, plus clear row-level error reporting.</li><li>• Staging table keeps raw data separate from the target table, so wrong data never directly corrupts live records.</li></ul>                                                                                                                                                       |
| 4.    | Social Impact / Customer<br>Satisfaction    | <ul style="list-style-type: none"><li>• Saves hours of manual data entry and reduces human error, so staff can focus on more useful work.</li><li>• Gives accurate and trustworthy data, which improves decision making and service quality.</li><li>• Simple enough for non-technical users, which increases customer satisfaction and adoption.</li><li>• Reduces stress and paper / rework, and supports digital transformation of small and large organizations.</li></ul>                                                                                                                                       |

| S.No. | Parameter                         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5.    | Business Model<br>(Revenue Model) | <ul style="list-style-type: none"> <li>• Subscription / licence model: offered as a feature or add-on of a platform or as a managed service.</li> <li>• Pay-per-use or tiered pricing based on the number of records imported or the number of users.</li> <li>• Paid services: implementation, custom transform map development, data migration and training.</li> <li>• Support and maintenance contracts for enterprise customers.</li> </ul>                           |
| 6.    | Scalability of the Solution       | <ul style="list-style-type: none"> <li>• Handles growing file sizes and record counts through batch / asynchronous transforms.</li> <li>• New Transform Maps and target tables can be added without redesigning the solution.</li> <li>• Cloud-hosted multi-tier architecture supports many concurrent users and scheduled imports.</li> <li>• Can be extended to other sources such as databases and REST APIs, and to features like predictive field mapping.</li> </ul> |